FrameContact: Contact-Derived Task Routing for Antibody–Antigen Interfaces

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Abstract

Antibody–antigen interface prediction couples three related outputs: residue-pair contacts and their antibody-side paratope and antigen-side epitope marginals. A single jointly trained checkpoint, however, can be suboptimal for one output. We introduce FrameContact, a partner-conditioned architecture that combines protein-language-model residue embeddings, sequence-predicted local backbone frames, an explicit low-rank contact map, and interleaved bidirectional cross-attention. At inference, antibody and antigen structures are predicted separately from sequence; no experimental complex coordinates, native complex pose, or cross-partner geometry are required. Under a fixed SAbDab2 development protocol, epoch-level diagnostics and a 2×2 task ablation show that the fully joint objective gives the strongest epitope result. The curves are consistent with late overfitting and do not support simple multitask competition. We therefore retain joint learning and change only the epitope readout: an epitope expert aggregates the two strongest predicted contact edges per antigen residue, and deterministic task routing uses this expert only for epitope while preserving contact and paratope from the joint baseline. Before full-development retraining, we freeze this route and evaluate its screen-stage checkpoints once on an antigen-cluster-disjoint development holdout of 156 complexes. All three seeds improve epitope MPC-AP (+0.0057, +0.0141, and +0.0089); a paired bootstrap over 49 antigen clusters gives a cluster-weighted, seed-averaged improvement of +0.0295 with 95% CI $[0.0126, 0.0474]$, conditional on the three trained seeds. We then retrain on the complete 2,565/549 development split using only sequence-predicted geometry as model input. The routed three-seed validation medians are 0.0267/0.2438/0.6885 for contact/epitope/paratope MPC-AP. Relative to matched bound-geometry development checkpoints, the medians change by $-0.0015/+0.0067/-0.0233$, respectively. This is a development sensitivity analysis: the 549-complex validation set selects checkpoints, and a cluster-weighted paired analysis does not establish an epitope improvement. The fresh156 result validates the earlier routing decision, not the subsequently retrained checkpoints. These results support task-specific routing while showing that deployment without experimental structure preserves contact and aggregate epitope ranking more readily than paratope ranking.

1 Introduction

An antibody–antigen interface can be represented by a sparse binary matrix $Y \in \{0,1\}^{L_{\mathrm{Ab}} \times L_{\mathrm{Ag}}}$, where $Y_{ij} = 1$ marks a residue-pair contact. Paratope and epitope labels are deterministic row and column marginals of the same contact matrix. This relation makes joint learning attractive, but it also creates a practical model-selection problem: antigen-residue ranking and preservation of pair-level contacts or antibody-side residues can favor different checkpoints.

Recent methods have made epitope prediction antibody-aware using bipartite graph learning, multimodal interaction estimation, or interleaved cross-attention [1, 6, 13]. We study these design choices under one frozen SAbDab2 protocol and ask: *does the epitope gap arise from early optimization, late overfitting, or competition with contact and paratope supervision?* We then ask whether an epitope-specific readout can improve antigen-side prediction while retaining the stronger contact and paratope behavior of the joint model.

Our contributions are:

- We formulate contact, paratope, and epitope prediction around one explicit residue-pair contact map, combining pretrained sequence features with local N/CA/C geometry predicted separately for each partner and interleaved bidirectional cross-attention. Experimental complex coordinates are not required at inference.
- A controlled 2×2 task ablation and epoch-level analysis produce curves consistent with late overfitting and do not support simple multitask competition, motivating us to retain the fully joint objective.

- We introduce a single-factor TopK2 epitope readout and deterministic task routing. The route improves epitope prediction on a previously unused, antigen-cluster-disjoint development holdout while leaving contact and paratope exactly unchanged.
- We bind data, candidate grids, partner panels, checkpoints, and seeds with immutable manifests across screening, a cluster-disjoint route confirmation, full-development retraining, and a disclosed non-blind historical measurement.
- On the complete development split, we replace bound-complex model input with sequence-predicted partner geometry and quantify the resulting task-specific trade-off without accessing the historical or temporal sets.

2 Related work

Antibody-aware interface prediction. AsEP provides a clustered benchmark and casts antibody-specific epitope prediction as bipartite link prediction; its WALLE baseline combines protein language models with structural graphs [6]. MIPE jointly predicts paratopes and epitopes using multimodal contrastive learning and an estimated interaction matrix [13]. EpiFormer interleaves bidirectional cross-attention inside geometric encoders and combines it with sparsity-aware objectives [1]. SPPIDER-seq demonstrates the value of partner-conditioned cross-attention in general protein-interface prediction [10]. Our EpiFormer-inspired component is a controlled readout adaptation inside our own 3D pair-contact architecture; the original EpiFormer and this adaptation have distinct model graphs.

Sequence and structure priors. Protein language models such as ESM-2 encode evolutionary regularities into residue features [5]. Geometry-aware graph networks provide useful inductive biases for molecular coordinates [12]. Paraplume shows that PLM representations alone can be strong and scalable for antigen-agnostic paratope prediction [2]. Our setting is instead partner-specific: antibody and antigen sequences are both inputs, and their chain-internal geometry is predicted before interface inference. This removes the need for an experimentally resolved complex at deployment. The completed experiment still uses experimental structures to derive contact targets and an auxiliary distance target during training, so we distinguish sequence-conditioned inference from a fully structure-free training pipeline.

3 Method

3.1 Geometry-aware pair-contact backbone

For each chain, external ESM-2 embeddings [5] are projected to $d = 128$. We construct a right-handed frame (R_i, o_i) at each residue from its N–Cα–C atoms, with o_i at Cα. For an ordered same-chain pair (i, j), the invariant geometric descriptor is

$$g_{ij} = \left[\mathrm{RBF}_{16}(\|o_j - o_i\|),\ R_i^\top \,\mathrm{unit}(o_j - o_i),\ \mathrm{vec}(R_i^\top R_j),\ \delta_{ij}^{\mathrm{sgn}}, \delta_{ij}^{\mathrm{abs}}\right], \tag{1}$$

where $\mathrm{unit}(0) = 0$ and the two sequence-offset features are clipped after normalization by 32. Each antibody or antigen block factorizes intra-chain encoding into (i) dense content self-attention with sequence rotary positions and (ii) self-attention whose per-head logits receive an MLP bias from g_{ij}. The geometric branch uses self edges plus at most 32 nearest same-chain neighbors within 16 Å; residues with missing or degenerate frames remain in the content path but contribute no learned geometric bias. The branches are fused as $H_{\mathrm{intra}} = H_{\mathrm{dense}} + \tanh(\gamma_{\mathrm{geo}})(H_{\mathrm{geom}} - H)$, where H is the block input and $\gamma_{\mathrm{geo}} = 0$ at initialization. Antibody and antigen use separate parameters.

For the deployment-style experiment reported below, backbone coordinates come from AlphaFold 3 predictions generated separately for the antibody and antigen components from their sequences, with templates disabled. We retain only each component's chain-internal N/CA/C geometry: no experimental coordinates, cross-chain relative coordinates, native complex pose, or true cross-chain distance is supplied as model input. Experimental bound structures are used to derive the supervised contact matrix and the auxiliary distance bins during training, but are not required at inference.

We alternate this intra-chain update with partner exchange rather than applying one late fusion step. At layer $\ell \in \{1, 2\}$, separate geometry-aware encoders first produce (A_ℓ, G_ℓ); the shared pair decoder forms a stage

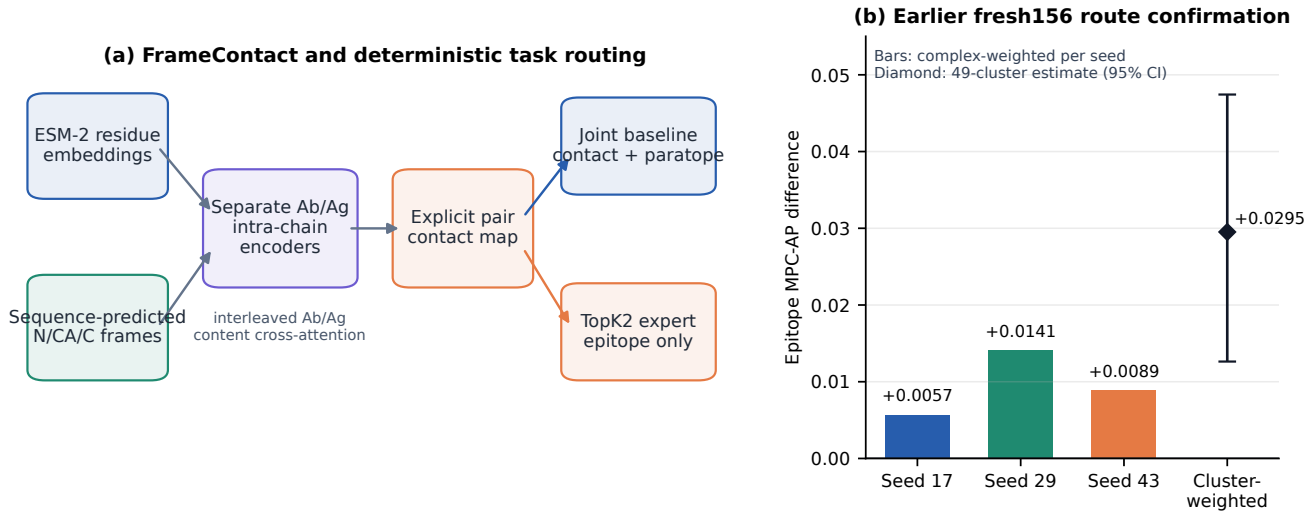


Figure 1: **Model and evidence flow.** FrameContact predicts an explicit residue-pair contact map from PLM embeddings and sequence-predicted intra-chain backbone geometry. The joint baseline supplies contact and paratope; the TopK2 expert supplies epitope only. The route is a fixed task-to-checkpoint map. Right: two estimands from the earlier bound-geometry fresh156 route confirmation—per-seed complex-weighted gains and the cluster-weighted, seed-averaged estimate with its conditional interval.

contact map; and bidirectional cross-attention updates both streams before layer $\ell+1$. Let h_i^{Ab} and h_j^{Ag} denote the embeddings at a given stage. The same directional low-rank decoder is applied in both chain orders:

$$\begin{aligned} \phi(x,y) &= \frac{q(x)^\top k(y)}{\sqrt{d_p}} + u(x) + v(y) + b, \\ z_{ij} &= \tfrac{1}{2}\left[\phi(h_i^{\mathrm{Ab}}, h_j^{\mathrm{Ag}}) + \phi(h_j^{\mathrm{Ag}}, h_i^{\mathrm{Ab}})\right], \quad p_{ij} = \sigma(z_{ij}), \end{aligned} \tag{2}$$

where $q, k : \mathbb{R}^d \to \mathbb{R}^{d_p}$ are learned projections, $u, v : \mathbb{R}^d \to \mathbb{R}$ are learned unary maps, $b \in \mathbb{R}$, and $d_p = 128$. The head is evaluated over the complete antibody–antigen Cartesian candidate grid. Antibody-to-antigen attention at layer ℓ uses content logits only:

$$a_{ij}^{h,\mathrm{Ab}\leftarrow\mathrm{Ag}} = \underset{j:M_{ij}=1}{\mathrm{softmax}}\left(\frac{(W_q^h h_i^{\mathrm{Ab}})^\top (W_k^h h_j^{\mathrm{Ag}})}{\sqrt{d_h}}\right), \tag{3}$$

with the analogous update in the reverse direction. Its output is $A'_\ell = A_\ell + \tanh(\gamma_\ell^{\mathrm{Ab}})\,\mathrm{CrossAttn}(A_\ell, G_\ell)$, and analogously for G'_ℓ; both cross-stream gates start at zero. The next intra-chain block consumes these partner-conditioned features. Thus, local SE(3)-invariant predicted-chain geometry and partner-specific content exchange are interleaved without supplying the native cross-chain pose. Although stage contact logits are exposed for diagnostics, the frozen configuration is `content_only`: no function of z_{ij} enters the attention logits.

Pair supervision is read from the frozen binary contact cohort. Residue labels are derived strictly from the pair matrix,

$$y_i^{\mathrm{para}} = \bigvee_j Y_{ij}, \qquad y_j^{\mathrm{epi}} = \bigvee_i Y_{ij}, \tag{4}$$

rather than imported surface annotations. In the reported joint baseline, paratope prediction combines a direct antibody-node logit with a masked log-mean-exp contact marginal, whereas epitope prediction is the logit of the masked mean pair probability:

$$\begin{aligned} s_i^{\mathrm{para}} &= w_p^\top \mathrm{LN}(h_i^{\mathrm{Ab}}) + \mathrm{LME}_{j:M_{ij}=1}(z_{ij}), \\ s_j^{\mathrm{epi,base}} &= \mathrm{logit}\left[\mathrm{clip}\left(\frac{1}{|I_j|}\sum_{i\in I_j} p_{ij}, \epsilon, 1-\epsilon\right)\right], \qquad I_j = \{i : M_{ij} = 1\}, \end{aligned} \tag{5}$$

where $\mathrm{LME}_{r\in S}(z_r) = \log \sum_{r\in S} \exp(z_r) - \log |S|$. Applying a sigmoid gives the residue probabilities. The joint loss is

$$\mathcal{L} = \mathcal{L}_c + 0.1\mathcal{L}_d + 0.1\mathcal{L}_p + 0.5\mathcal{L}_e, \qquad \mathcal{L}_e = \mathcal{L}^e_{\text{CB-BCE}} + \mathcal{L}^e_{\text{Dice}} + 0.2\mathcal{L}^e_{\text{prev}}. \tag{6}$$

For each complex, binary losses are averaged within each observed class, then equally across observed classes; complexes also receive equal weight. $\mathcal{L}_c$ uses focal BCE with $\gamma = 1$ [4], whereas $\mathcal{L}_p$ and $\mathcal{L}^e_{\text{CB-BCE}}$ use BCE. With $\varepsilon = 10^{-6}$, the epitope-specific terms are

$$\begin{aligned} \mathcal{L}^e_{\text{Dice}} &= \frac{1}{B}\sum_b \left[1 - \frac{2\sum_r p_{br} y_{br} + \varepsilon}{\sum_r p_{br} + \sum_r y_{br} + \varepsilon}\right], \\ \mathcal{L}^e_{\text{prev}} &= \frac{1}{B}\sum_b \left|\frac{1}{n_b}\sum_r p_{br} - \frac{1}{n_b}\sum_r y_{br}\right|. \end{aligned} \tag{7}$$

Here, B is the number of complexes in the minibatch, n_b is the number of valid antigen residues in complex b, and p_{br} and y_{br} are the predicted epitope probability and binary label for residue r. $\mathcal{L}^e_{\text{Dice}}$ follows the differentiable Dice objective [8]. $\mathcal{L}_d$ is class-balanced cross-entropy on Cα–Cα distances bucketed at $4, 8, 16$, and 32 Å into five classes; it is averaged equally over observed classes within each complex and is auxiliary only.

3.2 Contact-derived epitope expert

Mean-like marginalization can dilute the few high-confidence edges of a sparse interface. For antigen residue j, the TopK2 expert instead computes

$$\begin{aligned} I_j &= \{i : M_{ij} = 1\}, & k_j &= \min(2, |I_j|), \\ \mu_j &= \frac{1}{k_j}\sum_{i\in \mathrm{TopK}_{k_j}(\{p_{ij}: i\in I_j\})} p_{ij}, & s^{\text{epi}}_j &= \mathrm{logit}(\mathrm{clip}(\mu_j, \epsilon, 1-\epsilon)), \end{aligned} \tag{8}$$

where ϵ is float32 machine epsilon. Empty candidate sets map to zero logit, although the complete Cartesian grid prevents this case for real antigen residues. The only intentionally changed forward computation between the two training arms is the epitope readout. The backbone, contact decoder, supervision, loss weights, training data, candidate grid, and checkpoint-selection rule remain fixed. The TopK2 readout participates in training and supplies the epitope logits used by the epitope loss; it is not a post-hoc inference-only transformation.

3.3 Deterministic task routing

For each seed, we compose the outputs from two independently trained, seed-matched checkpoints:

$$\left(\hat{P}^c_r, \hat{p}^{\text{para}}_r, \hat{p}^{\text{epi}}_r\right) = \left(f^c_{\theta_{J,r}}(x), f^p_{\theta_{J,r}}(x), f^e_{\theta_{E,r}}(x)\right), \qquad r \in \{17, 29, 43\}, \tag{9}$$

where $\theta_{J,r}$ is the seed-matched joint baseline checkpoint and $\theta_{E,r}$ is the seed-matched TopK2 expert selected on development data. The route assigns one frozen checkpoint to each output using fixed task identity; the routed quantities are continuous scores, not thresholded labels.

4 Experimental protocol

Data and splits. We use a 3,663-complex SAbDab2 structural cohort [3] with a locally frozen, label-free antigen-fold split; it is not an official SAbDab2 split. Antigen components combine Foldseek reciprocal-min TM-score edges at 0.5 with exact PDB-assembly, antigen-sequence, and antibody-sequence identity; seed 20260730 assigns 2,565/549/549 complexes to train, validation, and the historical partition with zero exact-identity overlap across roles. Pair labels come from a frozen binary contact cache over the complete Cartesian candidate grid (190,509,766 pairs; 143,756 positives). The cache receipt authenticates the binary labels but does not independently record the upstream atom-level cutoff or missing-residue policy, so we do not infer either definition from the cache. For the 3,114 train/validation complexes used in the final sensitivity study, the antibody and antigen components are predicted separately from sequence and cached on the same canonical residue axes. Predicted structures for the historical partition are not materialized. All compared development arms use the same labels, candidate grid, external embeddings, and 31-way reciprocal partner panels.

Evidence stages. The 512/128 screen is used for task ablation and route design. Before measuring fresh156, we freeze the route and remove every instance and antigen cluster used by earlier selection or confirmation screens, leaving 156 complexes in 49 clusters. This measurement evaluates the frozen screen-stage route. We then retrain both arms on all 2,565 training complexes and select checkpoints on the 549-complex validation set. We first freeze the route under bound-geometry development, then repeat the same six seed-and-arm runs after replacing model input geometry with sequence-predicted partner structures. Consequently, fresh156 is independent evidence for the preceding routing decision, whereas the full-development and geometry-replacement tables are validation summaries, not independent confirmation. The previously exposed 549-complex partition is reported only as a measurement-only historical appendix result and is not used in the predicted-geometry study.

Optimization and selection. We use three seeds (17, 29, 43), bf16, two interleaved stages, eight heads, dropout 0.1, AdamW [7] with learning rate 3×10^{-4} and weight decay 10^{-4}, and batch size one. Checkpoints maximize validation epitope MPC-AP, break ties by contact MPC-AP and then earlier epoch, and use at most 20 epochs with patience 5. The full validation split supplies checkpoint selection, while fresh156 supplies the earlier antigen-cluster-disjoint route confirmation.

Metrics and statistics. We implement "AUPRC" as scikit-learn's non-interpolated average precision (AP; `average_precision_score`), not trapezoidal PR area [9]:

$$\mathrm{AP} = \sum_k (R_k - R_{k-1}) P_k. \tag{10}$$

Because positives are sparse and complex sizes vary, the primary metric for task t averages AP only over positive-eligible complexes $\mathcal{E}_t = \{n : \sum_\ell y_{n\ell}^{(t)} > 0\}$:

$$\text{MPC-AP}_t = \frac{1}{|\mathcal{E}_t|} \sum_{n \in \mathcal{E}_t} \mathrm{AP}(y_n, s_n), \tag{11}$$

with PR metrics preferred to ROC-only summaries under imbalance [11]. Pooled metrics retain all labeled examples, including those belonging to a zero-positive complex. Let δ_{rn} be the routed-minus-joint AP difference for seed r and complex n, and let C_g contain the complexes in antigen cluster g. We report both

$$\Delta_{\text{seed}-\text{med}} = \operatorname*{median}_r \frac{1}{N} \sum_n \delta_{rn}, \qquad \Delta_{\text{cluster}} = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|C_g|} \sum_{n \in C_g} \frac{1}{3} \sum_r \delta_{rn}. \tag{12}$$

For fresh156, the confidence interval for Δ_{cluster} is the 2.5/97.5 percentile interval from 10,000 paired bootstrap resamples of the 49 antigen clusters after first averaging the three fixed seeds. This interval is conditional on those trained seeds and does not include retraining uncertainty. The frozen promotion gate required median epitope gain ≥ 0.005, at least two positive seeds, minimum seed gain ≥ -0.01, and exact preservation of routed contact and paratope metrics.

5 Results

5.1 Joint learning remains beneficial

Both the same-protocol baseline and EpiFormer-inspired model peak early: the median best epoch across seeds is 2. Training loss continues to fall afterward while validation epitope AP declines. These curves are consistent with late overfitting, but do not by themselves identify a unique cause. The controlled task ablation in Table 1 supports retaining the fully joint objective, which is strongest; adding contact to epitope-only training is also mildly beneficial.

Table 1: Three-seed median epitope MPC-AP on the frozen 512/128 development screen. Architecture, split, and seed set are fixed.

Supervised outputs	Epitope MPC-AP
Epitope only	0.2502
Epitope + paratope	0.2493
Epitope + contact	0.2509
Epitope + paratope + contact	**0.2532**

5.2 Task routing transfers to an unseen development holdout

On fresh156, the routed model improves epitope MPC-AP for every seed (Table 2). The paired antigen-cluster bootstrap point estimate is +0.0295 with 95% CI $[0.0126, 0.0474]$ for the cluster-weighted, seed-averaged estimand in Eq. (12). Contact and paratope are exactly unchanged because those outputs are bound to the baseline checkpoint.

Table 2: Fresh156 epitope MPC-AP for the screen-stage checkpoints. The route was frozen before labels from this antigen-cluster-disjoint panel were measured; these are not evaluations of the later full-development retraining.

Seed	Joint baseline	Routed	Difference
17	0.1858	0.1915	+0.0057
29	0.1895	0.2036	+0.0141
43	0.1849	0.1938	+0.0089
Median difference			**+0.0089**

5.3 Sequence-predicted geometry development sensitivity

We repeat the complete 2,565/549 development protocol after replacing bound-complex model input with partner-separated AlphaFold 3 predictions. The model receives no experimental coordinates as input during training or inference, and no cross-partner geometry is constructed. Table 3 reports the three-seed medians on the 549-complex validation set. Because this set also selects each checkpoint, the comparison is a development sensitivity analysis, not an independent confirmation.

Table 3: Full-development validation MPC-AP with bound-complex versus sequence-predicted geometry as model input. The final column is the median of the three seed-matched differences, not the difference between displayed medians. Validation supplies checkpoint selection for both conditions.

Metric	Bound geometry	Predicted geometry	Paired difference
Contact	0.02858	0.02673	−0.00155
Epitope	0.23746	0.24378	+0.00666
Paratope	0.70475	0.68847	−0.02327

The predicted-minus-bound epitope differences for seeds 17, 29, and 43 are +0.00791, +0.00666, and −0.01301; contact differences are +0.00189, −0.00155, and −0.00432; and paratope differences are −0.01627, −0.02496, and −0.02327. Thus, the seed-level aggregate keeps contact close and has a positive median epitope change, but paratope declines consistently.

A complementary paired analysis first averages the three fixed seeds within each complex and then weights 108 antigen clusters equally. Its predicted-minus-bound estimates are −0.00247 for contact (95% bootstrap CI $[-0.00498, 0.00003]$), −0.00563 for epitope ($[-0.02016, 0.00880]$), and −0.02781 for paratope ($[-0.03570, -0.02045]$). The apparent epitope direction therefore depends on the estimand: the median of seed-level aggregate metrics is positive, whereas the cluster-weighted paired estimate is negative and its interval crosses zero.

6 Discussion and conclusion

The curves are consistent with late overfitting, and the task ablation does not support simple contact/paratope competition as an explanation for the epitope gap. Contact-derived TopK2 aggregation plus deterministic routing produces a consistent epitope gain on fresh156 under the earlier frozen route. Replacing experimental model-input geometry with sequence-predicted partner structures then preserves contact approximately and yields a positive median seed-level epitope change, but the cluster-weighted epitope interval crosses zero and paratope declines consistently. This mixed result is stronger evidence for deployment feasibility than for an overall performance improvement.

Cross-paper numbers from EpiFormer, Paraplume, and other published systems use different cohorts, splits, labels, and aggregation rules; we therefore do not treat them as a formal ranking. A defensible structural-method comparison requires rerunning each baseline on the identical split and candidate grid. The principal limitations are the absence of an independent evaluation of the final retrained checkpoints, the paratope sensitivity to predicted geometry, the use of bound-derived labels and auxiliary distance supervision during training, and the cache-level provenance boundary on the atom-level contact definition.

Reproducibility statement

Each checkpoint has 2,395,374 parameters. Routing evaluates two checkpoints, therefore requiring two forward passes. The expert reruns used one RTX 4090 per seed, bf16, batch size one, and 21.33 aggregate GPU-hours. Frozen ESM-2 input features use `facebook/esm2_t33_650M_UR50D`, revision `08e4846e5371`, 1,280 dimensions, and weight SHA-256 prefix `a08adabb949f`. The embedding-manifest SHA-256 prefix is `9ecbc0881516`.

The bound-geometry reference analyzer SHA-256 prefix is `f4368275e38d`. The predicted-geometry campaign and analyzer prefixes are `1117a6ac45bd` and `a4168a919e97`; its model-input structure manifest prefix is `bf6ea4a21ff3`. Full digests, split roles, and binary-contact authorities are stored in the accompanying machine-readable record. The accompanying machine-readable metrics file records the compact metrics, seeds, split roles, and audit status used in this draft. The archive also includes a machine-readable audit of every bibliographic record.

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A Historical SAbDab2 benchmark

For transparency, Table 4 reports a previously exposed, single-seed, measurement-only historical benchmark. This partition had already been accessed, and prior test feedback influenced later development. The present measurement did not select the route, model, checkpoint, threshold, or calibration, but it is not a fresh blind test and cannot confirm generalization. Thresholds were selected on development validation and were not refit here. Pooled metrics use all 549 complexes; per-complex AP excludes one zero-positive complex and therefore averages over 548 eligible complexes.

Table 4: Previously exposed, non-blind, single-seed historical measurement; not used for model selection. AP columns are non-interpolated average precision.

Task	AUROC	AP	F1	MCC	MPC-AP
Contact	0.9371	0.0121	0.0370	0.0519	0.0264
Epitope	0.6915	0.1411	0.2119	0.1564	0.2486
Paratope	0.9567	0.6567	0.6440	0.6097	0.7095